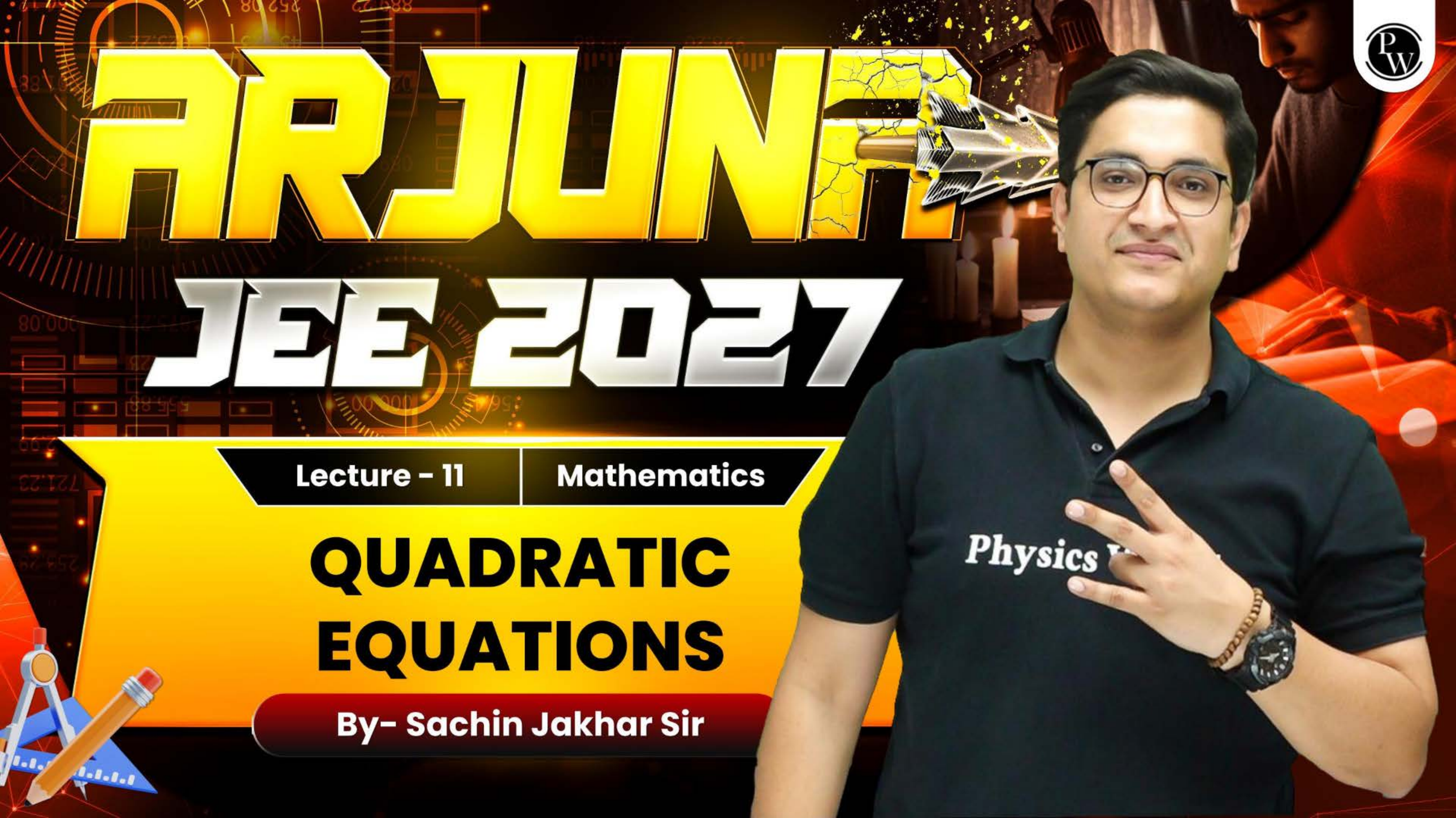
ARJUNA JEE 2027

Lecture - 11 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Common Root Condition**

B **Location of Roots**

C **Question Practice**

D



Question Practice

Critical Thinking Question (CTQ)

69%

If the range of the function $f(x) = \dfrac{x^2 + ax + b}{x^2 + 2x + 3}$ is $[-5, 4]$, $a, b \in N$, then find the value of $(a^2 + b^2)$

$14^2 + 9^2 = 196 + 81 = 277$

$y = \dfrac{x^2+ax+b}{x^2+2x+3}$ fnf

$x^2y + 2xy + 3y = x^2 + ax + b$

$(y-1)x^2 + (2y-a)x + (3y-b) = 0$

$\# \; D \geq 0$

$(2y-a)^2 - 4(y-1)(3y-b) \geq 0$

$4y^2 + a^2 - 4ay - 12y^2 + 12y + 4by - 4b \geq 0$

$-8y^2 + 4y(b+3-a) + a^2 - 4b \geq 0$

$$(2y-a)^2 - 4(y-1)(3y-b) \geq 0$$

$$4y^2 + a^2 - 4ay - 12y^2 + 12y + 4by - 4b \geq 0$$

$$-8y^2 + 4y(b+3-a) + a^2 - 4b \geq 0$$

$\ominus$

$$\boxed{8y^2 - 4(b+3-a)y + 4b - a^2 \leq 0}$$

Soln $\rightarrow$ y = Range.

$y \in [-5, 4]$



$\hookrightarrow y \in [-5, 4]$

$8y^2 - 4(b+3-a)y + 4b - a^2 = 0$ → -5, 4

$$\text{Sum} = -\frac{1}{8_2} = \frac{+(+4)(b+3-a)}{8_2}$$

$$-2 = b+3-a$$

$$a-b = 5$$

$$a-5 = b$$

$$14-5 = b$$

$$9 = b$$

$$\text{Prod.} = -20 = \frac{4b-a^2}{8}$$

$$-160 = 4b-a^2$$

$$-160 = 4(a-5)-a^2$$

$$-160 = 4a-20-a^2$$

$$a^2-4a-140 = 0$$

$$(a-14)(a+10) = 0$$

$a = 14, -10$ ✓, ✗

COMMON ROOT CONDITION

Case – I : If both roots are common

$a_1x^2 + b_1x + c_1 = 0$ — α, β

$a_2x^2 + b_2x + c_2 = 0$ — α, β

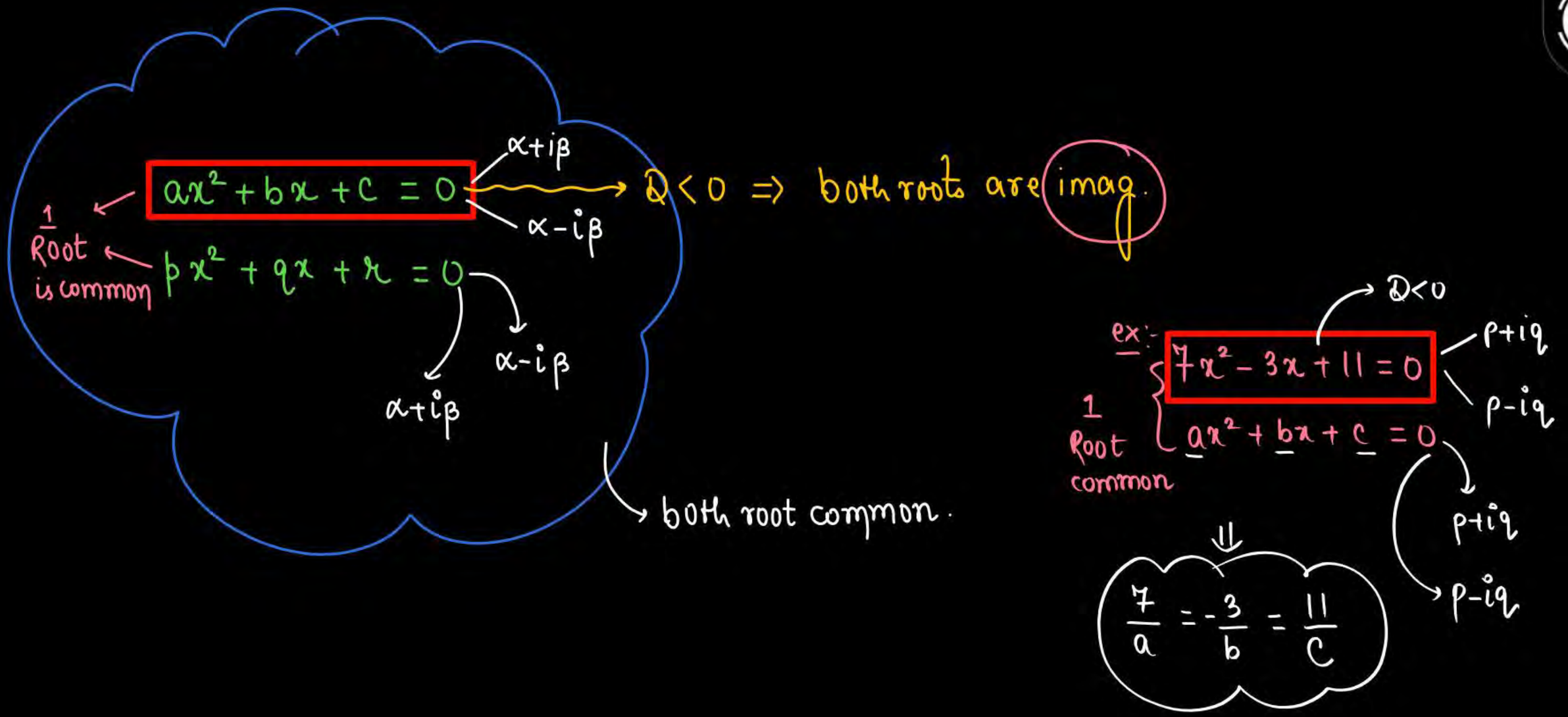
$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

ex:-

$x^2 - 11x + 28 = 0$ — 7, 4

$-3x$

$-3x^2 + 33x - 84 = 0$ — 7, 4


$ax^2+bx+c=0$
$\alpha+i\beta$
$\alpha-i\beta$
$D<0 \Rightarrow$ both roots are imag.
1 Root is common
$px^2+qx+r=0$
$\alpha-i\beta$
$\alpha+i\beta$
both root common.
ex:-
$7x^2-3x+11=0$
$D<0$
$p+iq$
$p-iq$
1 Root common
$ax^2+bx+c=0$
$p+iq$
$p-iq$
$\frac{7}{a}=\frac{-3}{b}=\frac{11}{c}$



COMMON ROOT CONDITION

Case – II : If atleast one root ~~are~~ is common

Direct Result :

$$a_1x^2 + b_1x + c_1 = 0$$
$$a_2x^2 + b_2x + c_2 = 0$$

Condition for atleast one common root:

$$(a_1b_2 - b_1a_2)(b_1c_2 - b_2c_1) = (a_1c_2 - a_2c_1)^2$$

$a_1x^2 + b_1x + c_1 = 0$

$a_2x^2 + b_2x + c_2 = 0$

$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \cdot \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}^2$

$(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (a_1c_2 - a_2c_1)^2$

M-2: **Procedure**

$a_1x^2 + b_1x + c_1 = 0$ → α, β

$a_2x^2 + b_2x + c_2 = 0$ → α, γ

done Quad. $\overset{9}{\text{M}}$ (make coeff. of x^2 same.

Subtract.

value of x obtained is common root (α)

common root (α) that will satisfy both Quad.

Question

Find the value of p if the equation $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common root(s).

M-1 Santosh

D.R

$3x^2 - 2x + p = 0$

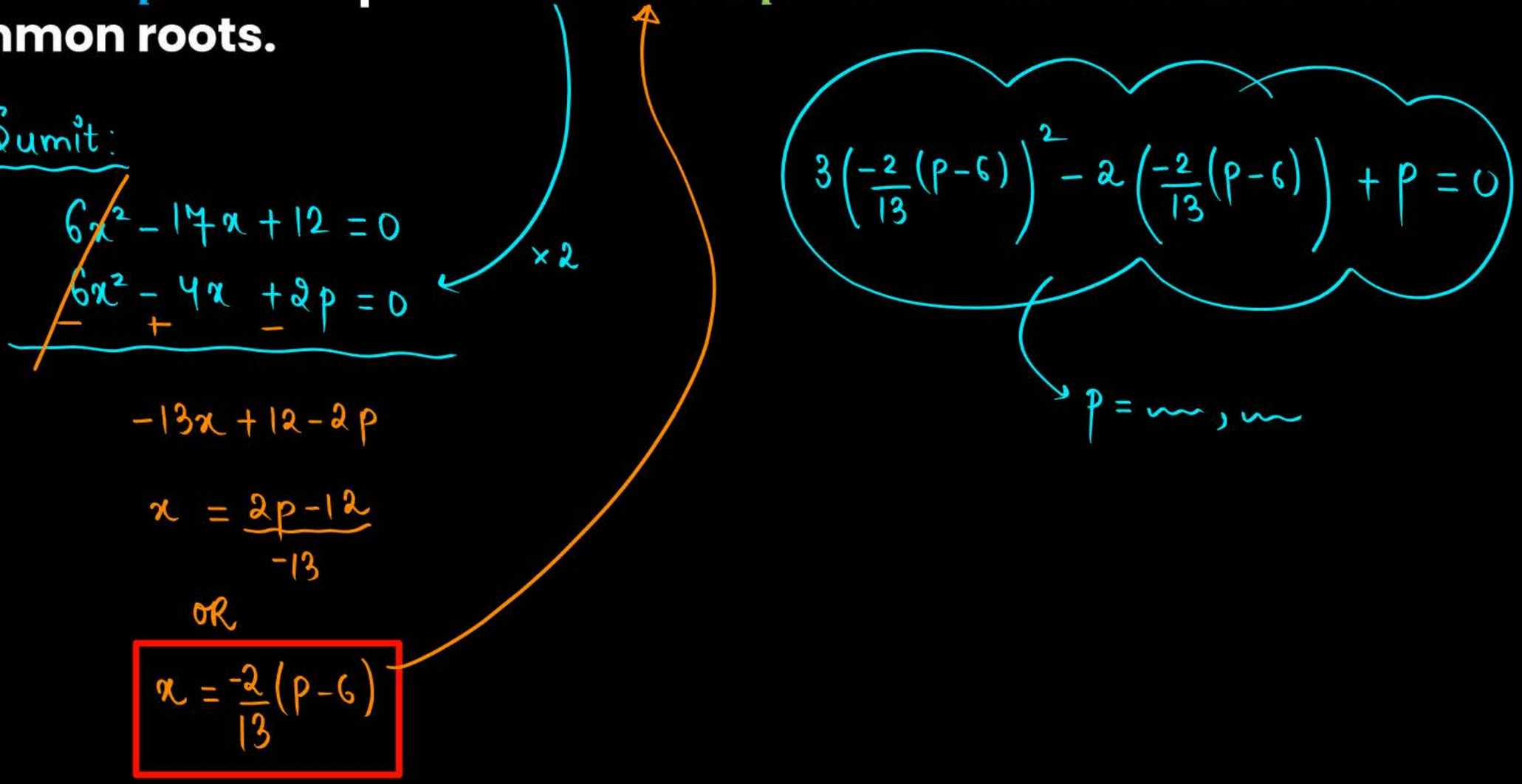
$6x^2 - 17x + 12 = 0$

$$\begin{vmatrix} 3 & -2 \\ 6 & -17 \end{vmatrix} \times \begin{vmatrix} -2 & p \\ -17 & 12 \end{vmatrix} = \begin{vmatrix} 3 & p \\ 6 & 12 \end{vmatrix}^2$$

$$\left(-3\times17 - (-12)\right)\left(-24 - p\times(-17)\right) = (36 - 6p)^2$$

Question

Find the value of p if the equation $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common roots.



Question

Find the value of p if the equation $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common roots.

$6x^2 - 9x - 8x + 12 = 0$

$(3x-4)(2x-3) = 0$

$\# \; x = \frac{3}{2}, \frac{4}{3}$

(M-3) Sunita.

Creative Method

$x = \frac{3}{2} \Rightarrow 3 \times \frac{9}{4} - \cancel{2} \times \frac{3}{\cancel{2}} + p = 0$

$p = 3 - \frac{27}{4} = \frac{-15}{4}$

$x = \frac{4}{3} \Rightarrow \cancel{3} \times \frac{16}{\cancel{9}_3} - 2 \times \frac{4}{3} + p = 0$

$p = \frac{8}{3} - \frac{16}{3} = \frac{-8}{3}$

Question [JEE Mains-2023]

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real roots is $\frac{3}{\sqrt{2\beta}}$, then β is equal to

77%

$x^2 - 5ax + 1 = 0$

$x^2 - ax - 5 = 0$

$$\begin{vmatrix} 1 & -5a \\ 1 & -a \end{vmatrix} \times \begin{vmatrix} -5a & 1 \\ -a & -5 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -5 \end{vmatrix}^2$$

$$(-a-(-5a)) \times (25a+a) = (-5-1)^2$$

$$4a \times 26a = 36$$

$$a^2 = \frac{9}{26}$$

$$a = \frac{\pm 3}{\sqrt{26}}$$

$2\beta = 26$

$\beta = 13$ ✓

Question [JEE Mains-2013]

If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common roots, then $a : b : c$ is

$p+iq$
$p-iq$

$p+iq$, $p-iq$.

$D < 0$

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{3}$$

A 1 : 2 : 3

B 3 : 2 : 1

C 1 : 3 : 2

D 3 : 1 : 2

Acidic Question

The equations $kx^2 + x + k = 0$ and $kx^2 + kx + 1 = 0$ have exactly one root in common for

A $k = -1/2, 1$

B $k = 1$

C $k = -1/2$

D $k = 1/2$

$kx^2 + x + k = 0$

$kx^2 + kx + 1 = 0$

$$\begin{vmatrix} k & 1 \\ k & k \end{vmatrix} \cdot \begin{vmatrix} 1 & k \\ k & 1 \end{vmatrix} = \begin{vmatrix} k & k \\ k & 1 \end{vmatrix}^2$$

$$(k^2 - k)(1 - k^2) = (k - k^2)^2$$

$$0 = (k - k^2)^2 + (k - k^2)(1 - k^2)$$

$$(k - k^2)(k - k^2 + 1 - k^2) = 0$$

$$-k(1-k)(2k+1)(k-1) = 0$$

$k = 0, 1, -\frac{1}{2}$

Apply atleast 1 Root common cond^n ↓ Check !!

Acidic Question

The equations $kx^2 + x + k = 0$ and $kx^2 + kx + 1 = 0$ have exactly one root in common for

A $k = -1/2, 1$

B $k = 1$

C $k = -1/2$

D $k = 1/2$

for $k=0$ ✗

$x = 0, \quad 1 = 0$

$k = 1$ ✗

$x^2 + x + 1 = 0$

$x^2 + x + 1 = 0$ → both root Common.

$k = 0, 1, -\frac{1}{2}$

$k = -\frac{1}{2}$

$-\frac{x^2}{2} + x - \frac{1}{2} = 0$

$-x^2 + 2x - 1 = 0$

$x^2 - 2x + 1 = 0$ (1, 1)

$x^2 + x - 2 = 0$ (1, −2)

Apply atleast 1 Root common cond^n ↓ Check !!

Acidic Question

The equations $kx^2 + x + k = 0$ and $kx^2 + kx + 1 = 0$ have exactly one root in common for

A $k = -1/2, 1$

B $k = 1$

C $k = -1/2$

D $k = 1/2$

$kx^2 + x + k = 0$

$kx^2 + kx + 1 = 0$

$x - kx + k - 1 = 0$

$k - 1 = (k-1)x$

$0 = (k-1)x - (k-1)$

$k = 1$

$0 = (k-1)(x-1)$

$x = 1$

C.R.

$x = 1$

$k + 1 + k = 0$

$k = -\frac{1}{2}$

Apply atleast 1 Root common cond^n ↓ Check !!

Acidic Question

If the quadratic equation $x^2 + bx + c = 0$ & $x^2 + cx + b = 0$ $(b \neq c)$ have a common root, then prove that their uncommon roots are the roots of the equation $x^2 + x + bc = 0$

β
γ

$x^2 + bx + c = 0$ — $\alpha = 1$, β

$x^2 + cx + b = 0$ — $\alpha = 1$, γ

$(b-c)x + c - b = 0$

$\# \quad (b-c)x = b-c$

$x = 1 \rightarrow$ C.R.

$1 + \beta = -b$

$\beta = c$

$1 + \gamma = -c$

$\gamma = b$

$1 + b + c = 0$

$b + c = -1$

$x^2 - (\beta+\gamma)x + \beta\gamma = 0$ — γ, β

$x^2 - (c+b)x + cb = 0$

-1

$x^2 + x + bc = 0$ ✓

Hence.

Acidic Question

If the quadratic equations $x^2 + ax + 12 = 0$ and $x^2 + bx + 15 = 0$ and $x^2 + (a + b)x + 36 = 0$ have a common positive root find 'a' and 'b' and the root of the equation.



Question [AIEEE-2008]

The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio $4 : 3$. Then the common root is



A 1

B 4

C 3

D 2

If $x = \alpha$ is Common Root of

$P_1(x) = 0$ → α

$P_2(x) = 0$ → α

$$m\,P_1(x) \pm n\,P_2(x) = 0 \rightarrow \alpha$$

m, n → Const. / variable.

$P_1(x)$

ex:- $x^3 - 6x^2 + 11x - 6 = 0$ → 1, 2, 3

$x^2 - 5x + 6 = 0$ → 2, 3 ($P_2(x)$)

$\times x$ →

$$x^3 - 6x^2 + 11x - 6 = 0$$
$$x^3 - 5x^2 + 6x = 0$$

→ $P_1(x) - xP_2(x) = 0$

$$-x^2 + 5x - 6 = 0$$

$x^2 - 5x + 6 = 0$ → 2, 3

$$7(x^3 - 6x^2 + 11x - 6) - 3(x^2 - 5x + 6) = 0 \quad \text{→ } 2, 3$$

$$7(x^3 - 6x^2 + 11x - 6) - x(x^2 - 5x + 6) = 0 \quad \text{→ } 2, 3$$

Question

If cubic $2x^3 + bx^2 + cx + d = 0$ (α, β, γ) & quadratic $x^2 + px + q = 0$ (α, β) have two roots in common, then show that: $\left((b - 2p) = \frac{c-2q}{p} = \frac{d}{q}\right)$

$2x^3 + 2px^2 + 2qx = 0$

$2x^3 + bx^2 + cx + d = 0$

$-\quad -\quad -\quad -$

HHSP

$(2p-b)x^2 + (2q-c)x - d = 0$ — roots α, β

$1x^2 + px + q = 0$ — roots α, β

$\frac{2p-b}{1} = \frac{2q-c}{p} = -\frac{d}{q}$

Question

Let $f(x) = x^3 - 3x + b$ and $g(x) = x^2 + bx - 3$, where b is a real number. If the equation $f(x) = 0$ and $g(x) = 0$ have a common root, then

Multiple Correct.



A Number of possible values of b is 3.

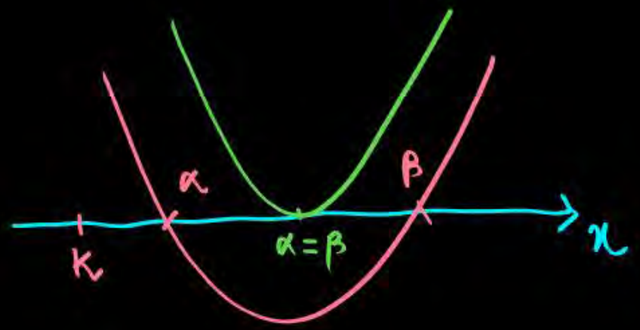
B Number of possible values of b is 2.

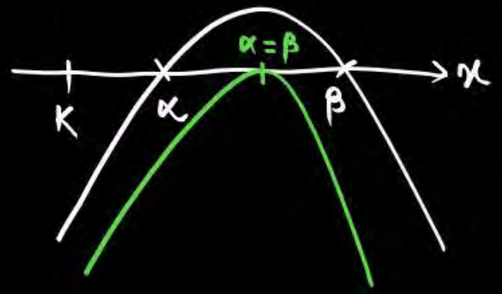
C Sum of all possible values of b is 0.

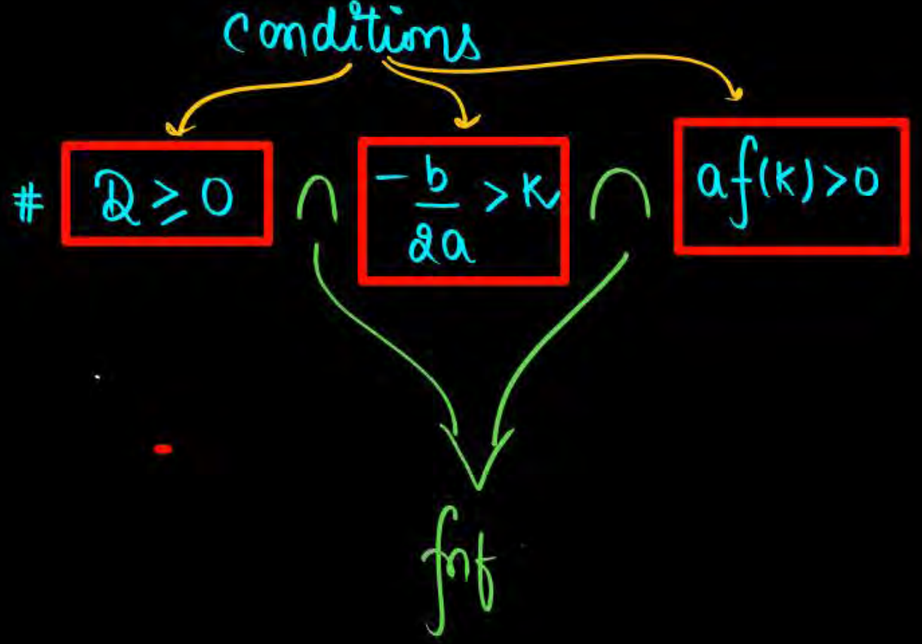
D Sum of all possible values of b is 2.

LOCATION OF ROOTS

Case-I : If both roots are greater than number given number 'k'

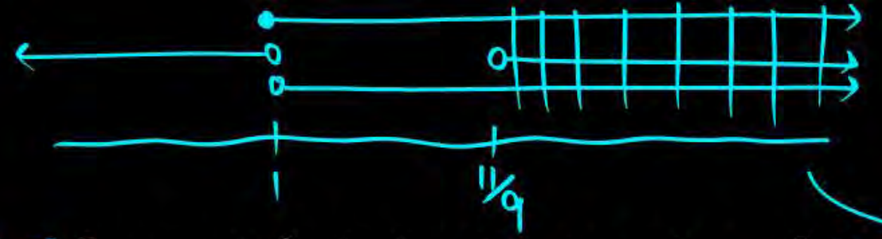






Question

Find all the values of the parameter 'd' for which both roots of the equation $x^2 - 6dx + (2 - 2d + 9d^2) = 0$ exceed the number 3.



greater

k

Cond^n:

$D \geq 0$

$36d^2 - 4(2 - 2d + 9d^2) \geq 0$

$9d^2 - (2 - 2d + 9d^2) \geq 0$

$9d^2 - 2 + 2d - 9d^2 \geq 0$

$2(d - 1) \geq 0 \Rightarrow d \geq 1$



$\frac{-b}{2a} > 3$

$\frac{+(+6d)}{2 \times 1} > 3$

$d > 1$

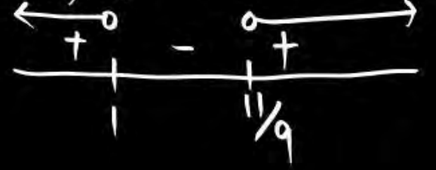
$af(3) > 0$

$1 \cdot (9 - 18d + 2 - 2d + 9d^2) > 0$

$9d^2 - 20d + 11 > 0$

$9d^2 - 9d - 11d + 11 > 0$

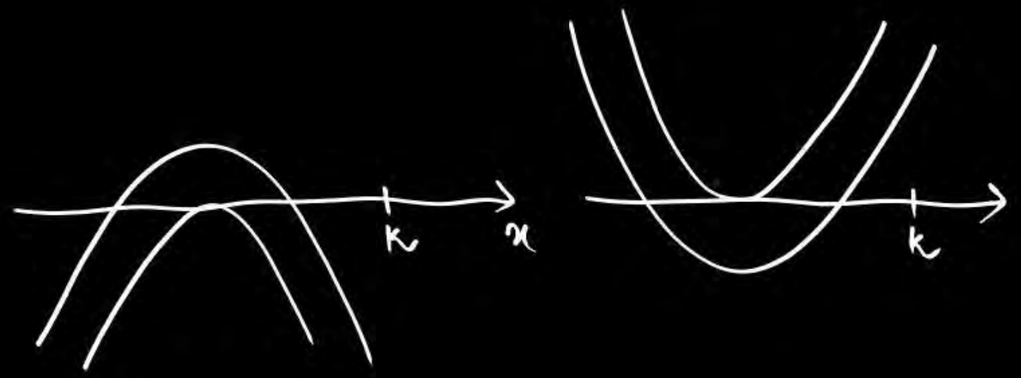
$(9d - 11)(d - 1) > 0$



$d \in \left(\frac{11}{9}, \infty\right)$

LOCATION OF ROOTS

Case-II : If both roots are less than number given number 'k'



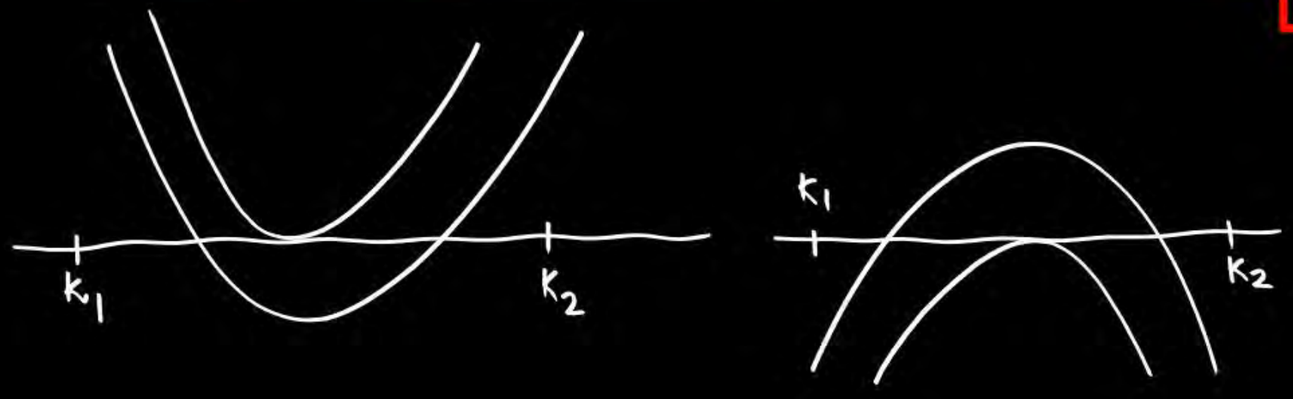
cond^{n}.

$D \geq 0 \cap \frac{-b}{2a} < k \cap a.f(k) > 0$

LOCATION OF ROOTS

Case-III : Both roots lies between 'k_1' & 'k_2'.

Case-III : (C-I ∩ C-II)



Cond^{n}.

$D \geq 0 \cap k_1 < -\frac{b}{2a} < k_2 \cap af(k_1) > 0 \cap af(k_2) > 0$

Question

If α, β are the roots of the quadratic equation $x^2 + 2(k-3)x + 9 = 0$ $(\alpha \neq \beta)$.
If $\alpha, \beta \in (-6, 1)$, then find the values of k.

DO IT BY YOURSELF (DIBY)

RANGE OF RATIONAL FUNCTION

DIBY-47

We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \leq p$ is satisfied for any real x. Find the smallest integral good number.

[Ans. : 4]

DIBY-48

If α, β are the roots of equation $x^2 + x - 2 = 0$, then find the value of

$$\frac{\alpha\beta^4(\beta+1)^4+\beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}.$$

DIBY-49

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

[Ans. : 6]

DIBY-50

The number of real solutions of the equation $3\left(\overbrace{x^2+\frac{1}{x^2}}^{t^2-2}\right)-2\left(\overbrace{x+\frac{1}{x}}^{t}\right)+5=0$, is

[JEE Mains-2023]

(A) 4 (B) 0 (C) 3 (D) 2

$x+\frac{1}{x}=t$

$x^2+\frac{1}{x^2}+2=t^2$

[Ans. : (B)]

COMMON ROOT CONDITION

DIBY-51

If $x^2+4ax+3=0$ and $2x^2+3ax-9=0$ have a common root then the value of $|a|$ is

(A) 1 (B) 0 (C) -1 (D) None of these

[Ans. : A]

DIBY-52

If a, b, p, q are non-zero real numbers, then the equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have:

(A) No common root

(B) One common root if $2a^2 + b^2 = p^2 + q^2$

(C) Two common roots if $3pq = 2ab$

(D) Two common roots if $3qb = 2ap$

[Ans. : A]

DIBY-53

$x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

(A) 24 (B) 0, 24 (C) 3, 24 (D) 0, 3

[Ans. : B]

DIBY-54

If $(x^2 + px + 1)$ is a factor of $(ax^3 + bx + c)$, then

(A) $a^2 + c^2 = -ab$ (B) $a^2 - c^2 = -ab$

(C) $a^2 - c^2 = ab$ (D) None of these

[Ans. : C]

DIBY-55

The value of p and q for which the equation $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots is common.

(A) $p = \frac{3}{2}, q = 4$ (B) $p = 2, q = \frac{3}{4}$ (C) $p = \frac{2}{3}, q = 4$ (D) $p = 4, q = \frac{3}{2}$

[Ans. : (A)]

DIBY-56

If the equation $x^2 - 4x + 5 = 0$ and $x^2 + ax + b = 0$ (where $a, b \in \mathbb{R}$) have a common roots then which of the following is correct?

(A) $a = -4, b = -5$ (B) $a + b = 1$

(C) $a + b = -1$ (D) $b - a = 1$

[Ans. : (B)]

DIBY-57 ✳ ✳ ✳

If $Q_1(x) = x^2 + (k - 29)x - k$ and $Q_2(x) = 2x^2 + (2k - 43)x + k$ both are factors of a cubic polynomial $P(x)$, then the largest value of k is

(A) 0 (B) 33 (C) 23 (D) 30

[Ans. : (D)]

DIBY-58

If the equation $x^2 + abx + c = 0$ and $x^2 + acx + b = 0 (b \neq c)$ have a common root then their uncommon roots are the roots of the equation

(A) $a(b+c)x^2 + (b+c)x + abc = 0$ (B) $a(b+c)x^2 - (b+c)x - abc = 0$

(C) $a(b+c)x^2 - (b+c)x + abc = 0$ (D) $a(b+c)x^2 + (b+c)x - abc = 0$

[Ans. : (D)]

DIBY-59

If $(a-b)x^2 + 2(b-c)x + (a-c+1) = 0 (a, b, c \in \mathbb{R})$ and $x^2 + 2x + 5 = 0$ have a common roots, then which of the following is true?

(A) $a - b = 1$ (B) $a - b = \frac{1}{3}$ (C) $a - b = \frac{1}{2}$ (D) $a - b = 2$

[Ans. : (B)]

DIBY-60

If equations $ax^2 - (a+b)x + b = 0$ & $bx^2 + (b-c)x - c = 0$ has exactly one root in common $\{a, b, c \neq 0\}$, then which of the following can be correct?

(A) $b^2 = ac$ (B) $-a = b \neq c$ (C) $b = a \neq c$ (D) $-a \neq b = c$

[Ans. : A, B, D]

DIBY-61

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

[JEE MAINS - 2020]

(A) 18 (B) 36 (C) 9 (D) 27

[Ans. : (A)]

DIBY-62

If one root is square of the other root of equation $x^2 + px + q = 0$, then the relation between p and q is

[IIT-JEE-2004]

(A) $p^3 - q(3p - 1) + q^2 = 0$

(B) $p^3 - q(3p + 1) + q^2 = 0$

(C) $p^3 + q(3p - 1) + q^2 = 0$

(D) $p^3 + q(3p + 1) + q^2 = 0$

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: (3 sittings of 1 HR each)

Exercise-02(Prabal): 1 to 10,13,16,18 to 25.

Exercise-03(Setu): 1 to 30 (except-13,16,17,21,25,26,29)

It is never about end result

It's always about Journey

Thank You

#futureIITians